

POKHARA UNIVERSITY

Level: Master

Semester: Spring

Year : 2026

Programme: ME/MSc Computer

Full Marks : 100

Course: Database Engineering (New)

Pass Marks : 60

Time : 4 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the characteristics of a relational model. Discuss the concepts of domain, tuple, attribute, relation schema, and relational constraints with suitable examples. 7
- b) Explain CAP theorem. Compare different Architectural models for distributed DBMS. 8
2. a) Given Relational Schema 10

- Employee(eid, emp_name, address, supervisor_id)
- Department(did, dep_name)
- Project(pid, prj_name, did)
- Workson(eid, pid, hours)

Write relational algebraic expression and SQL query for following operations.

- List the name of all employee for computer department along with their supervisor Name.
- Find the name of all employees who work on network monitoring project for more than 15 hrs.
- Delete all the project belonging to computer department.
- Find the total number of projects from each department along with department name.
- Find the employee's name who do not work in any project.

- b) Explain relational algebra joins in detail. 5
3. a) Explain the working principles of B-Tree and B+ Tree indexes. Compare their structure, searching process, and suitability for database applications. 7

OR

5. Explain materialized views and discuss different techniques for maintaining their freshness. How can materialized views improve the performance of complex analytical queries?

6. b) A distributed database stores data across multiple geographical sites. Explain how fragmentation and replication can be used to improve performance, availability, and data locality, and discuss one major challenge of each technique.

4. a) Explain the concept of schema flexibility in NoSQL databases. Discuss how NoSQL systems handle changing and semi-structured data compared with relational databases.

6. b) Consider a MongoDB collection containing student records with fields such as StudentID, Name, Program, and Marks. Write suitable MongoDB commands to perform basic data insertion, retrieval, modification, and deletion, and explain how these operations differ from traditional SQL-based operations.

5. a) A publishing company produces books on various subjects. The books are written by authors who specialize in one particular subject. The company employs editors who, not necessarily being specialists in particular area, each take sole responsibility for editing one or more book publications. Shop owner buys books from the publisher. Shop owner can buy many books but one book can be bought by one shop owner only. Books are uniquely identified by bookid. Construct a clean and concise ER diagram for the Book database.

b) Explain the concept of a decentralized database using blockchain technology. Discuss blocks, hashing, distributed consensus, and immutability, and explain their significance for maintaining trustworthy records.

OR

5. A large dataset contains sales records from multiple stores. How HDFS works and also Explain how the MapReduce programming model can be used to calculate total sales for each store, describing the Map and Reduce phases with a suitable example.

3. 6. a) Compare Redis and Memcached as database caching solutions. For a high-traffic web application, analyze which caching strategy you would choose and justify your choice based on persistence, data structures, scalability, fault tolerance, and performance.

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- b) Explain the role of vector databases in AI/ML applications such as semantic search and Retrieval-Augmented Generation (RAG). Discuss the major ethical concerns associated with modern database systems, including data privacy, bias, security, data ownership, and responsible use of AI-generated data. 8
7. Write short notes on: (Any two) 2×5
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- a) Consistency Models
 - b) Types of NoSQL Database
 - c) Database Partitioning